

SAN JUAN DE MARCONA, Peru

WMO: 847210

Lat: 15.358S

Lon: 75.135W

Elev: 197

StdP: 14.59

Time Zone: -5.00 (W05)

Period: 82-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	56.4	57.0	50.5	54.8	60.8	51.4	56.7	60.7	27.9	64.6	25.6	64.9	11.6	180	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.7	82.3	73.0	80.6	71.0	79.1	70.1	74.3	80.1	73.3	78.1	72.2	77.5	18.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	119.0	78.4	71.5	117.0	78.0	70.0	111.2	76.6	38.1	81.0	37.0	78.3	36.1	78.0	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	24.3	22.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.0	73.7	74.5	73.7	71.1	67.5	64.8	62.6	61.8	62.8	64.9	67.7	71.2
	DBStd	5.04	2.31	2.38	2.29	2.56	2.80	3.23	2.19	1.89	1.74	1.60	2.01	2.48
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	322	0	0	0	0	4	40	82	102	70	22	2	0
	CDD50	6566	734	685	734	632	544	444	391	366	384	462	532	658
	CDD65	1413	269	265	269	182	83	34	8	3	4	19	84	193
	CDH74	3709	867	1106	917	374	63	24	2	2	2	0	29	323
	CDH80	213	45	96	52	7	1	2	0	0	0	0	1	9
Wind	WSAvg	14.8	14.7	15.2	15.4	15.2	14.7	14.2	13.4	14.5	15.4	15.5	15.0	14.7
Precipitation	PrecAvg	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	0.6	0.4	0.0	0.0	0.1	0.0	0.1	0.4	0.6	0.3	0.2	0.2	0.1
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.7	84.5	82.8	80.8	78.5	75.6	73.1	70.0	70.2	72.0	76.6	80.9
		MCWB	74.2	74.7	73.7	70.9	69.6	66.7	63.7	62.2	62.4	63.3	67.8	71.3
	2%	DB	80.9	82.5	81.4	79.1	75.3	73.4	69.5	68.1	68.3	71.1	74.6	78.8
		MCWB	71.5	73.4	72.1	70.0	67.3	65.3	62.2	60.8	61.1	63.0	66.2	70.2
	5%	DB	79.2	80.6	79.6	77.4	73.5	71.4	67.7	66.5	67.5	69.9	73.2	77.0
		MCWB	70.3	71.2	70.0	68.6	65.6	64.6	61.0	59.6	60.6	62.5	65.6	68.9
	10%	DB	78.0	79.1	78.5	75.9	71.9	69.5	66.1	65.1	66.3	68.9	71.8	75.5
		MCWB	69.8	70.4	69.3	67.6	64.8	63.0	60.0	59.0	60.0	62.0	64.8	68.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.1	75.7	75.4	72.8	71.1	67.8	65.2	62.9	63.7	65.6	69.0	72.4
		MCDB	82.4	82.7	81.3	77.9	76.0	74.2	70.5	68.7	69.3	70.1	74.7	79.2
	2%	WB	73.6	74.5	74.1	71.6	69.0	66.3	63.5	61.3	62.2	64.1	67.5	71.1
		MCDB	78.8	80.6	78.9	77.5	73.4	72.4	68.0	66.1	66.7	69.7	72.6	77.3
	5%	WB	72.4	73.5	72.8	70.2	67.5	65.0	62.0	60.6	61.2	63.1	66.6	69.9
		MCDB	77.7	78.6	77.4	75.7	72.3	70.9	66.2	65.2	65.6	68.5	71.5	75.3
	10%	WB	71.1	72.3	71.5	68.9	65.7	63.6	60.6	59.6	60.6	62.4	65.6	69.0
		MCDB	76.6	77.6	77.0	74.7	71.1	68.9	65.0	64.1	64.9	67.6	70.5	74.3
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.7	10.6	10.3	9.2	8.7	8.0	7.8	8.2	8.6	8.9	9.4
		MCDBR	10.1	11.4	11.1	11.0	10.2	10.0	9.8	9.2	9.3	9.4	10.1	10.0
		MCWBR	4.7	5.1	5.1	5.7	5.0	5.0	5.2	4.8	4.7	4.8	4.8	4.8
	5% WB	MCDBR	9.3	9.8	9.6	10.2	9.1	9.5	9.1	8.8	8.7	9.0	9.5	9.6
		MCWBR	5.0	4.9	5.1	5.9	5.3	5.3	5.6	5.3	4.9	5.1	5.1	5.1
Clear-Sky Solar Irradiance	taub	0.455	0.465	0.439	0.465	0.414	0.419	0.406	0.410	0.451	0.438	0.409	0.435	
	taud	2.275	2.266	2.353	2.216	2.304	2.263	2.310	2.294	2.157	2.207	2.323	2.285	
	Ebn at Noon	284	278	279	259	263	255	263	272	271	282	295	289	
	Edh at Noon	46	46	41	44	38	38	37	40	49	48	43	45	
All-Sky Solar Radiation	RadAvg	2274	2165	2125	1919	1635	1327	1275	1498	1809	2156	2376	2346	
	RadStd	137	141	80	83	95	84	76	84	100	91	111	113	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page